3/19/2019 Ch 5 : Compression System

Objective : To provide desired static enthalpy (ψ) as efficiently as possible. Would also like to :

- provide proper mass flux ($\dot{m}_a$)
- have flow in correct direction
- Provide correct combustor inlet Mach #
- Operate efficiently over a wide range of Mach # & angle of attack
- Isolate upstream shock train behavior from back pressure fluctuations.
- Minimize spillage & ext. drag.

□ η_c will be our primary measure of compression efficiency. Due to shocks & viscous losses $\eta_c < 1$.

□ Entropy generation.

Let $\beta \equiv$ shock strength $\equiv \dfrac{p_2 - p_1}{p_1} = \dfrac{p_2}{p_1} - 1$

Normal shock eq^n

$$\beta = \left(\frac{2\gamma}{\gamma+1}\right)\left(M_1^2 - 1\right)$$

↳ for oblique shock, use M_{1n} ← normal to shock

© North Carolina State University, All Rights Reserved

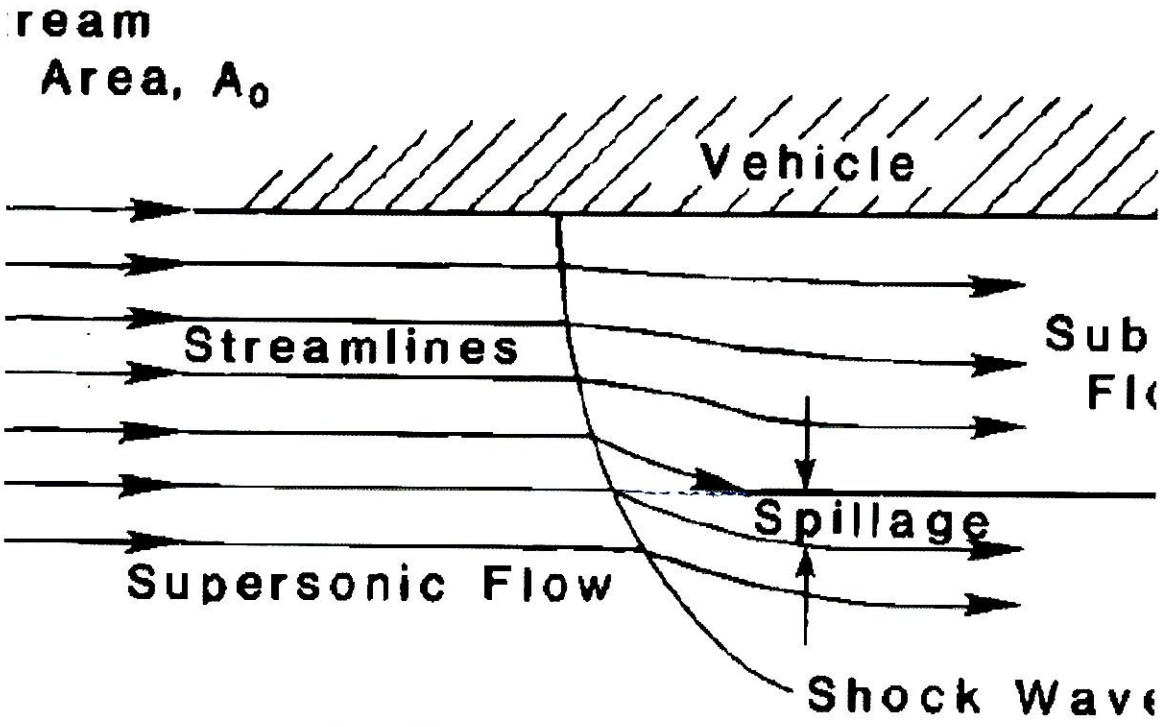

ream
Area, A_0
Vehicle
Streamlines
Sub
Fl
Spillage
Supersonic Flow
Shock Wave

Fig 5.1

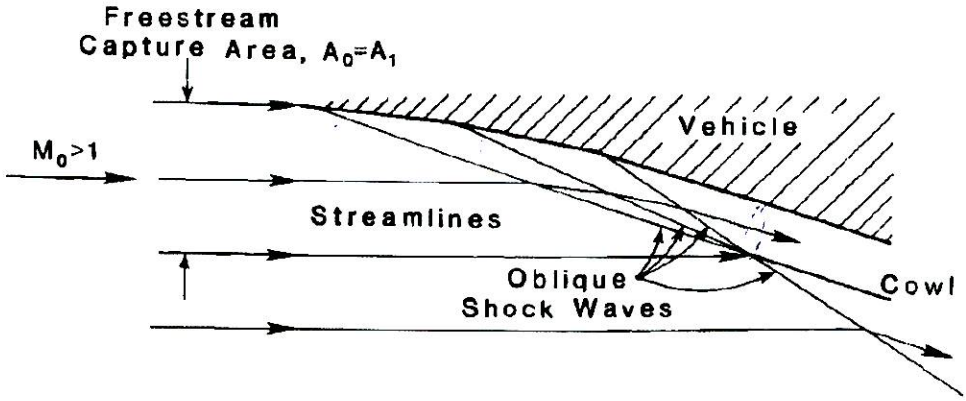


Fig. 5.2 External compression or oblique shock wave compression system. This device is shown at its design point, for which the several oblique shock waves generated by the external compression surface are "focused" on and terminate at the edge or lip of the engine cowl. The internal flow downstream of the final or "terminal" oblique shock wave is uniform and parallel. The uncaptured external flow must adjust its direction due to the presence of the cowl by means of a single oblique shock wave having the same total turning angle as the flow entering the engine.

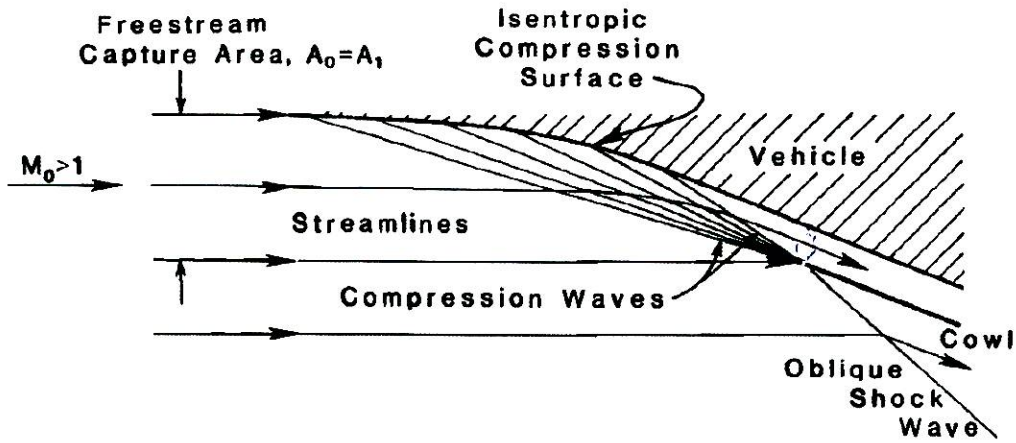


Fig. 5.3 Isentropic external compression system. This device is identical to that of Fig. 5.2, except that the compression surface is continuously curved in such a way that it generates a train of infinitesimal compression waves that cumulatively result in a finite pressure increase, but with no increase in entropy. The uncaptured external flow, however, must still adjust its direction by means of an oblique shock wave.

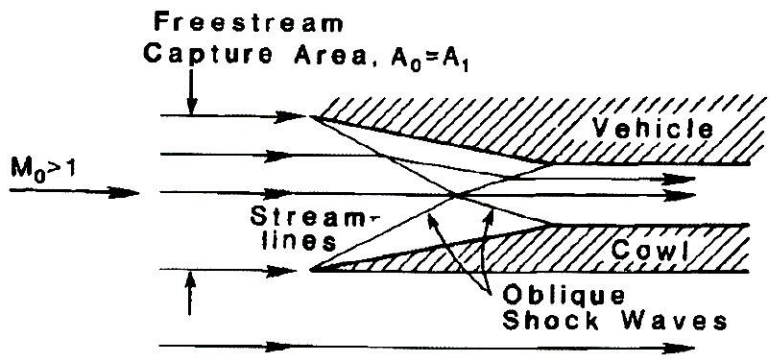


Fig. 5.5 Symmetrical internal compression system. This method employs an adjacent surface in order to generate a "mirror-image" oblique shock wave configuration that produces a uniform and parallel internal flow. This approach obviously shortens the axial length required for a given amount of compression, but requires an additional compression surface and can produce complex flows when operating away from the design point. The axisymmetric version of this type of compression system is known as the "Busemann inlet."

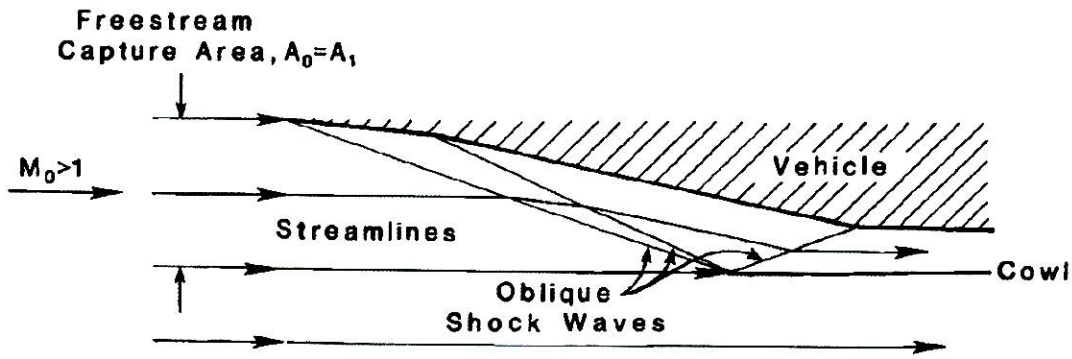
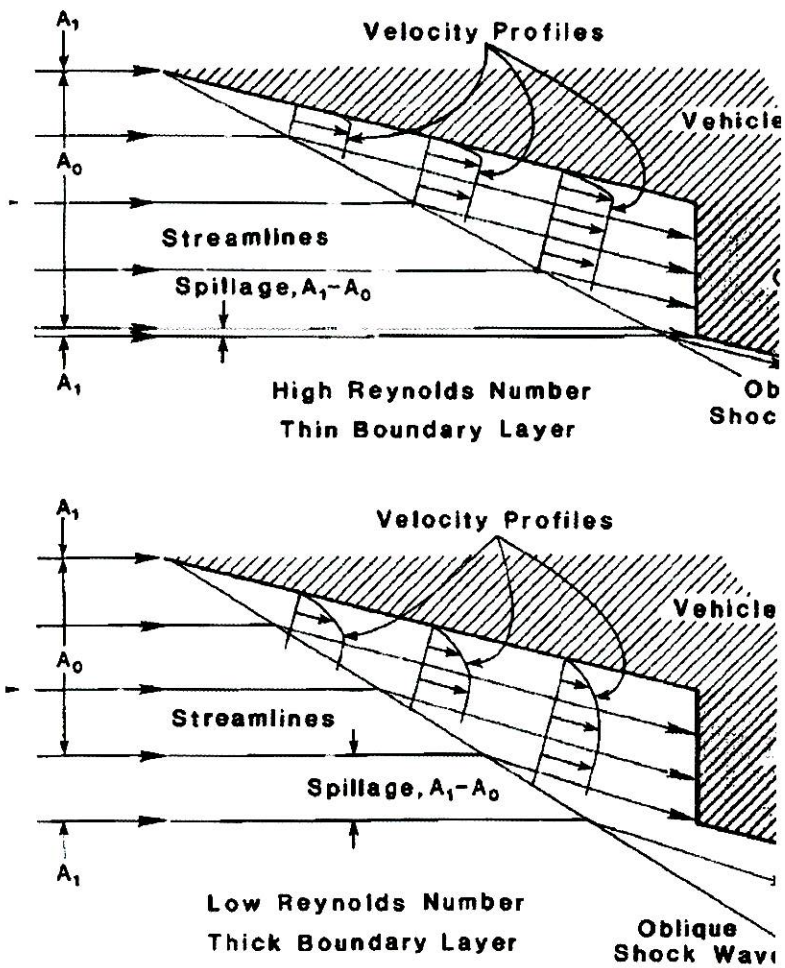


Fig. 5.4 Mixed external and internal compression system. This device incorporates oblique shock waves in various external and internal blends in order to achieve the desired combinations of compression and turning. In comparison to the external compression system of Fig. 5.2, this approach can use multiple internal reflections of weaker shock waves in order to accomplish the same task with less entropy increase, but the overall length must be greater. Mixed compression systems "decouple" the engine cowl angle from the amount of compression and can result in a cowl that is parallel to the freestream flow.



.7 Schematic representation of the behavior ace vehicle boundary layer upstream of the physic :he inlet.

From 2nd law of thermodyn.

$$\Delta s = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{p_2}{p_1}\right)$$

We can write $\frac{p_2}{p_1}$ & $\frac{T_2}{T_1}$ in terms of β

$$\Delta s = C_v \ln\left\{(1+\beta)\left[\frac{2\gamma + (\gamma-1)\beta}{2\gamma + (\gamma+1)\beta}\right]\right\}$$

For weak shocks.

$$\Delta s \sim R \frac{2}{3} \frac{\gamma}{(\gamma+1)^2} \underbrace{\left(M_1^2 - 1\right)}_{(M_{1n} \text{ for O.S})}^3$$

→ Δs is highly non-linear with M_1

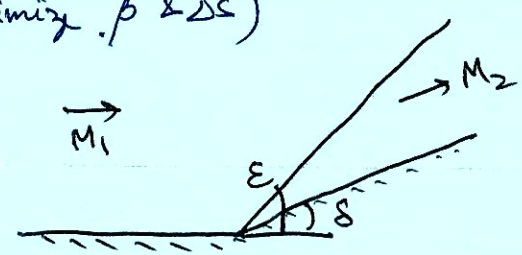
→ So we want to maximize η_c, so we need to minimize Δs, so we work to minimize M_{1n}.

→ Do this using oblique shocks (& minimize β & Δs)

To compare the change in Δs with N.S., we use hypothetical upstream Mach #, M_1'

$$M_1' \equiv M_1 \sin \varepsilon$$

If $\varepsilon = \frac{\pi}{2} \Rightarrow$ normal shock.



© North Carolina State University, All Rights Reserved

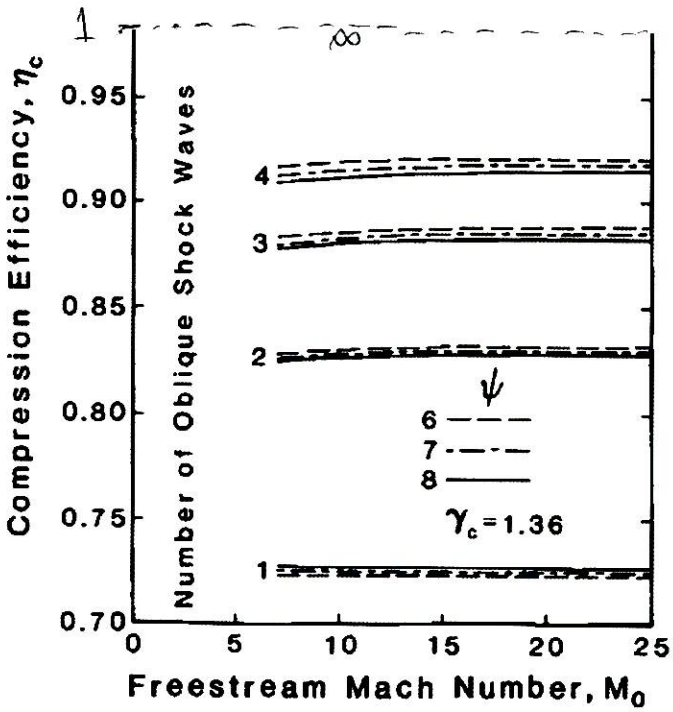


Fig. 5.16 Adiabatic compression efficiency as a function of freestream Mach number, static temperature ratio, and number

ε = functn of δ & M_1

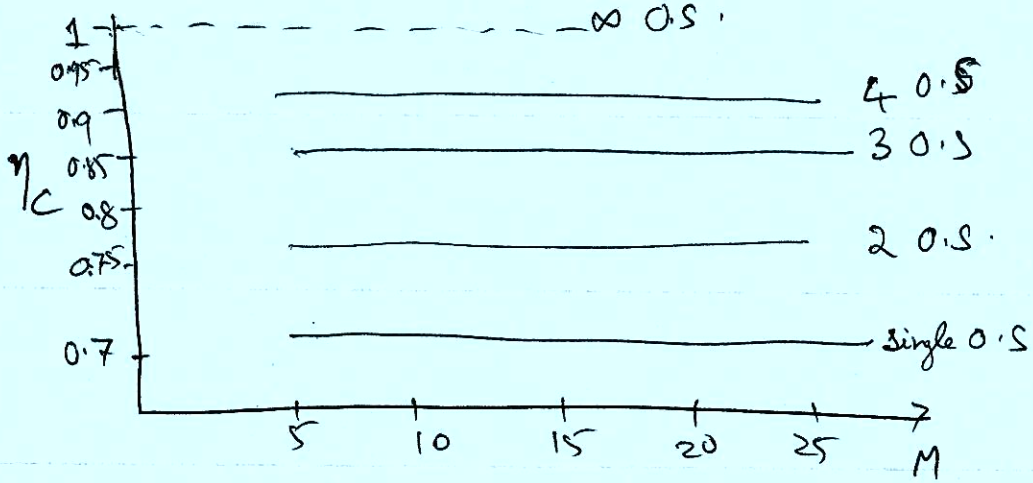
For small δ, set a small ε, $\Rightarrow$ small M'

$\Rightarrow$ low shock strength, small ΔS, high η_c

→ Will need to use more than one oblique shock to get sufficient compression

→ due to non-linear dependence of ΔS on Mach #, it is better to use a bunch of small shocks than 1 or 2 strong shocks.

→ ΔS will also be a functn of the compression ratio, which is measure by ψ



© North Carolina State University, All Rights Reserved

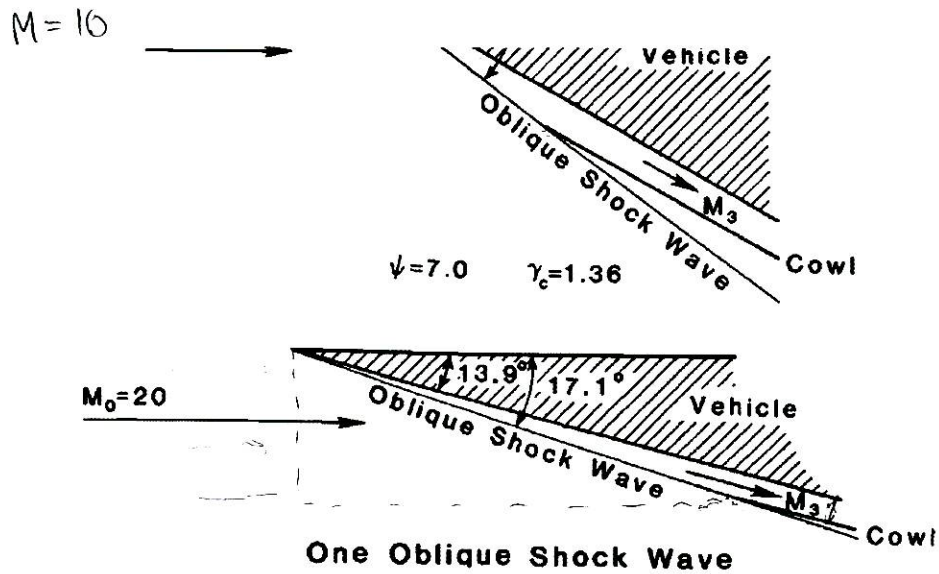


Fig. 5.11 Two external compression systems similar to that of Fig. 5.2, but incorporating *one* oblique shock wave. The vehicle wedge angle and the oblique shock wave angle both diminish with freestream Mach number because the static temperature ratio across an oblique shock wave is proportional to the square of the component of Mach number perpendicular to the wave front. [5.6]

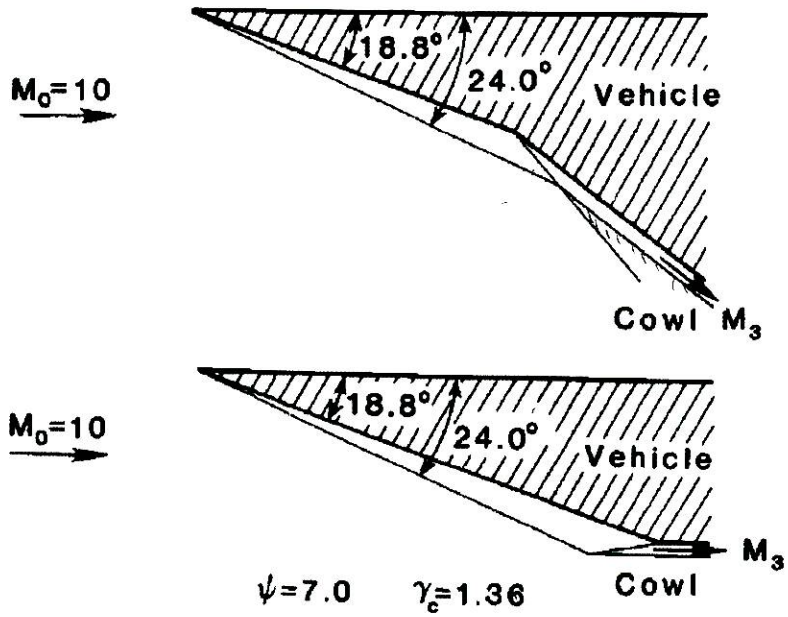


Fig. 5.12 Two compression systems that incorporate *two* oblique shock waves. The upper version is an external compression system, similar to that of Fig. 5.2, while the lower version is a mixed external and internal compression system, similar to that of Fig. 5.4. The lower version has the advantage of providing a cowl that is parallel to the freestream flow and therefore does not disturb the freestream flow, but is longer than the upper version.

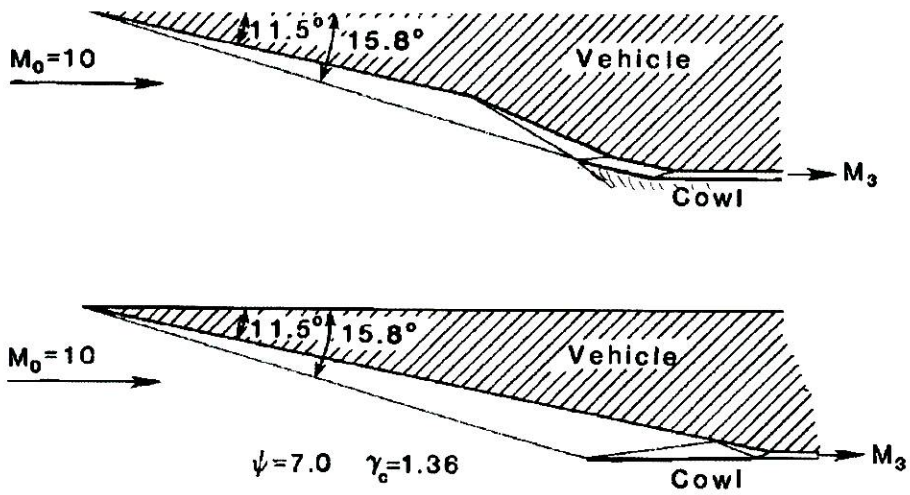


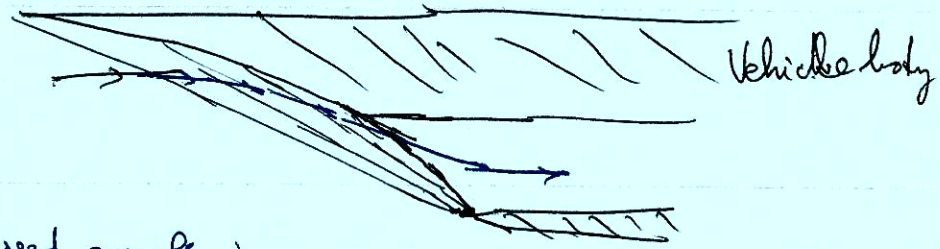
Fig. 5.14 Two mixed external and internal compression systems that incorporate *four* oblique shock waves. Both turn the combustor entrance flow parallel to the freestream, but the upper version, which has more external compression, will generate an oblique shock wave in the freestream because of the drooped cowl lip. The

To be effective, we want $\eta_c \simeq 0.9$ or better.

$\Rightarrow$ need to have an inlet that generates ≥ 4 shocks.

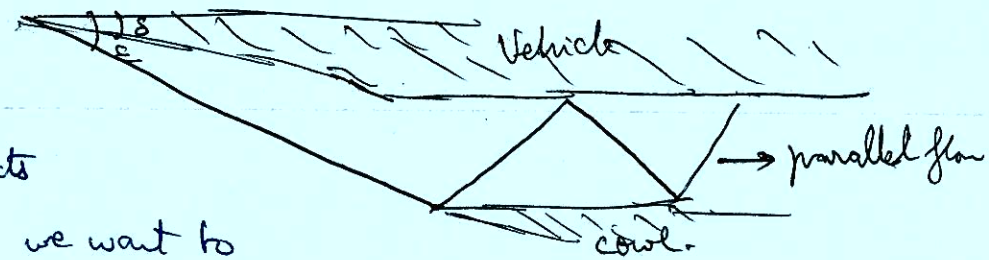
This can be done externally or with a combination of external & internal shocks

<u>All external</u>



- Oblique shock emanates from each turn of the flow.
- Would like to have the compressed gas flowing parallel to the internal geometry at the end of the shock waves.

<u>Ext & internal</u>



- Very good design on paper. But requires a long cowl (viscous effects in real engines are important & so we want to minimize the surface area available)

→ Want the leading oblique shock to intersect cowl lip to eliminate spillage.

→ However $\varepsilon = f(\delta, M_0)$

δ is fixed $\Rightarrow$ need to make a compromise & choose δ such that ε is reasonably well optimized for a range of Mach #s.

©North Carolina State University, All Rights Reserved

Compression Efficiencies

$$\eta_c = \frac{h_{3,s} - h_0}{h_3 - h_0}$$

In book $\eta_c' = \dfrac{h_3 - h_x}{h_3 - h_0}$

$$\frac{p_3}{p_0} = \left\{ \eta_c (\psi - 1) + 1 \right\}^{\gamma_c/(\gamma_c - 1)}$$

And we can also express η_c in terms of stag. pr. ratio Π_c

$$\Pi_c = \frac{p_{t3}}{p_{t0}} = \frac{p_3}{p_0} \left(\frac{\left(1 + \frac{\gamma - 1}{2} M_3^2\right)}{\left(1 + \frac{\gamma - 1}{2} M_0^2\right)} \right)^{\frac{\gamma_c}{\gamma_c - 1}}$$

But $T_{t3} = T_{t0}$ (adiabatic o.s.)

$$\Rightarrow \frac{T_{t3}}{T_{t0}} = \frac{T_3}{T_0} \left(\frac{1 + \frac{\gamma - 1}{2} M_3^2}{1 + \frac{\gamma - 1}{2} M_0^2} \right) \Rightarrow \psi = \frac{1 + \frac{\gamma - 1}{2} M_0^2}{1 + \frac{\gamma - 1}{2} M_3^2}$$

$$\Rightarrow \frac{p_{t3}}{p_{t0}} = \Pi_c = \frac{p_3}{p_0} \left(\frac{1}{\psi} \right)^{\gamma_c/(\gamma_c - 1)}$$

© North Carolina State University, All Rights Reserved

$$\frac{p_3}{p_0} = \pi_c \left(\psi\right)^{\left(\frac{\gamma_c}{\gamma_c - 1}\right)}$$

$$\left(\eta_c(\psi - 1) + 1\right)^{\frac{\gamma_c}{\gamma_c - 1}} = \pi_c \left(\psi\right)^{\frac{\gamma_c}{\gamma_c - 1}}$$

Solve for η_c

$$\boxed{\eta_c = \frac{\psi \left(\pi_c\right)^{\frac{\gamma_c - 1}{\gamma_c}} - 1}{\psi - 1}}$$

In book

$$\eta_c' = \frac{\psi - \left(\pi_c\right)^{-\left(\frac{\gamma_c - 1}{\gamma_c}\right)}}{\psi - 1}$$

Note $\eta_c' > \eta_c$

Gibbs eq^n

$$\Delta S = S_3 - S_0 = C_{p,c} \ln \underbrace{\left(\frac{T_3}{T_0}\right)}_{\psi} - R_c \ln \underbrace{\left(\frac{p_3}{p_0}\right)}_{f(\eta_c, \psi)}$$

$$\frac{\Delta S}{C_{p,c}} = \ln \left(\frac{\psi}{\eta_c(\psi - 1) + 1}\right)$$

or $\eta_c = \dfrac{\psi \exp\left(-\frac{\Delta S}{C_{p,c}}\right) - 1}{\psi - 1}$; in book $\eta_c' = \dfrac{\psi - \exp\left(\frac{-\Delta S}{C_{p,c}}\right)}{\psi - 1}$

© North Carolina State University, All Rights Reserved